

*Academic Excellence Through Quality Education*  
**KATHMANDU MODEL SECONDARY SCHOOL**  
 Second Term Examination- 2080

XI 'Science'  
 25 mins

Students are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

SET-A

Subject: Com. Mathematics (0071)

**GROUP 'A' MCQ [11×1=11]**

1. Choose the correct option of each question in your answer sheet.

2. If A and B are two sets then  $A \cap (A \cup B)^c$  is equal to

- a) B                      b)  $\emptyset$                       c)  $A - B$

3.  $|x + 2| \leq 1$  is equivalent to

- a)  $1 \leq x \leq \frac{1}{2}$                       b)  $1 \leq x \leq \frac{1}{3}$                       c)  $-1 \leq x \leq -\frac{1}{3}$                       d)  $-\frac{1}{2} \leq x \leq \frac{1}{3}$

4. A function  $f: N \rightarrow N$  is defined by  $f(x) = 2x + 1$ , for all  $x \in N$ , then f is

- a) one to one but not onto function                      b) onto but not one to one function  
 c) one to one and onto both function                      d) neither one to one nor onto function

5. The sum of the infinite Geometric series:  $(-\frac{5}{4} + \frac{5}{16} - \frac{5}{64} + \dots)$  is

- a)  $-\frac{1}{4}$                       b)  $\frac{5}{4}$                       c)  $-\frac{1}{2}$                       d)  $-\frac{1}{4}$

6. The distance between the lines  $2x - 5y - 6 = 0$  and  $6x - 15y + 11 = 0$  is

- a) 17                      b)  $\frac{29}{\sqrt{261}}$                       c)  $\frac{17}{\sqrt{261}}$                       d)  $\frac{17}{\sqrt{261}}$

7.  $\frac{|2-x|}{x-2}$  is

- a) 1                      b) -1                      c) doesn't exist                      d) doesn't exist

8. If  $y = \frac{1}{x}$  then  $\frac{dy}{dx}$  is

- a)  $\frac{y}{x}$                       b)  $-\frac{y}{x}$                       c)  $-\frac{y}{x}$                       d)  $-\frac{x}{y}$

9. For what value of 'a' the function  $f(x) = \begin{cases} ax - 5 & \text{if } x > 2 \\ x - 1 & \text{if } x \leq 2 \end{cases}$  is continuous at  $x = 2$ ?

- a) -2                      b) 1                      c) 2                      d) 3

10. The value of  $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{\ln(1+x)}$  is

- a) 1                      b) 2                      c) doesn't exist                      d) doesn't exist

11. The value of  $(1+i)^4 (1+\frac{1}{i})^4$  is

- a) 8                      b) 16                      c) 32                      d) 32

12. The lines given by  $ax^2 + 2hxy + by^2 = 0$  are real and distinct if

- a)  $h^2 > ab$                       b)  $h^2 < ab$                       c)  $h^2 = ab$                       d)  $a + b = 0$

Note: The question set must be written in the answer book.

11. The oxide which cannot act as a reducing agent is

- a)  $NO_2$                       b)  $CO_2$                       c)  $SO_2$                       d)  $ClO_2$

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**KATHMANDU MODEL SECONDARY SCHOOL**  
 Second Term Examination- 2080

Class: XI 'Science'

SET-B

Subject: Com. Mathematics (0071)

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

**GROUP 'A' MCQ [11×1=11]**

1. Choose the correct option of each question in your answer sheet.

2. If  $A = (-1, 3)$  and  $B = [0, 4]$  then  $A - B$  is

- a)  $(-1, 0)$                       b)  $[-1, 0)$                       c)  $(-1, 0)$                       d)  $[-1, 0]$

3. A function  $f: [-2, 2] \rightarrow R$ , defined by  $f(x) = x^2$  is

- a) One to one but not onto function                      b) onto but not one to one function  
 c) One to one and onto both function                      d) neither one to one nor onto function

4. The simplest form of  $\left(\frac{1-i}{1+i}\right)^3$  is

- a) 0                      b) 1                      c)  $3i$                       d)  $i$

5. The value of  $\lim_{x \rightarrow 0} \frac{\sin|x|}{x}$  is

- a) 0                      b) 1                      c) -1                      d) doesn't exist

6. The derivative of  $f(x) = (3-x)^5$  is

- a)  $5(3-x)^4$                       b)  $(3-x)^4$                       c)  $-5(3-x)^4$                       d)  $5(3-x)^5$

7. If A and B are two sets then  $A \cap (B - A)$  is equal to

- a) A                      b) B                      c)  $\emptyset$                       d) none

8. The sum of the infinite Geometric series:  $(-\frac{5}{4} + \frac{5}{16} - \frac{5}{64} + \dots)$  is

- a)  $\frac{1}{4}$                       b)  $-\frac{1}{4}$                       c)  $\frac{5}{4}$                       d) -1

9. The lines given by  $ax^2 + 2hxy + by^2 = 0$  are real and distinct if

- a)  $h^2 > ab$                       b)  $h^2 < ab$                       c)  $h^2 = ab$                       d)  $a + b = 0$

10. The distance between the lines  $2x - 5y - 6 = 0$  and  $6x - 15y + 11 = 0$  is

- a) 29                      b) 17                      c)  $\frac{29}{\sqrt{261}}$                       d)  $\frac{17}{\sqrt{261}}$

11. For what value of 'a' the function  $f(x) = \begin{cases} ax - 5 & \text{if } x > 2 \\ x - 1 & \text{if } x \leq 2 \end{cases}$  is continuous at  $x = 2$ ?

- a) 4                      b) -2                      c) 1                      d) 3

12. The value of  $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{\ln(1+x)}$  is

- a) 0                      b) 1                      c) 2                      d) doesn't exist

Note: The question set must be written in the answer book.

**KATHMANDU MODEL SECONDARY SCHOOL**  
Second Term Examination- 2080

XI 'Science'  
3 hrs.

Subject: Com. Mathematics (0071)

F.M.: 75

P.M.: 30

Answers are required to give their answers in their own words as far as practicable.  
Answers in the margin indicate full marks.

**GROUP 'B' [8×5=40]**

Define power set with example. [1+1+3]

For any two sets A and B

Prove that  $A - B = \overline{B} - \overline{A}$

For any sets A, B and C

Prove that  $A - (B \cap C) = (A - B) \cup (A \cap C)$

Define absolute value of real number. Interpret it geometrically. If 'a' be

positive real number and  $x \in \mathbb{R}$ ,

prove that:  $|x| \leq a \iff -a \leq x \leq a$ . [1+1+3]

Define one to one function.

If  $f: \mathbb{R} - \{3\} \rightarrow \mathbb{R} - \{1\}$  given by  $f(x) = \frac{x}{x-3}$ .

Show that f is bijective function. Also find  $f^{-1}(x)$  [1+3+1]

Define continuity of a function at a given point.

Function defined by

$$f(x) = \begin{cases} x^2 + 2 & \text{for } x < 5 \\ 20 & \text{for } x = 5 \\ 3x + 12 & \text{for } x > 5 \end{cases}$$

Does the limit of  $f(x)$  exist at  $x = 5$ ?

Is the function  $f(x)$  continuous at  $x = 5$ ?

If not, state the type of discontinuity? [1+1+1+1+1]

How can you make it continuous?

Two equation of straight lines are given by

$$L_1: a_1x + b_1y + c_1 = 0$$

$$L_2: a_2x + b_2y + c_2 = 0$$

Write the condition that the above lines  $L_1$  and  $L_2$  are perpendicular. [1]

Write the expression for the perpendicular distance from point [1]

$(x_1, y_1)$  to line  $L_1$  [3]

If  $z$  and  $w$  are two complex number, then prove that:

$$|z + w|^2 = |z|^2 + |w|^2 + 2\operatorname{Re}(z \cdot \overline{w})$$
 [2]

$$\text{Evaluate: } \lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x} - \sqrt[3]{1-x}}{x}$$
 [3]

$$\text{Evaluate: } \lim_{y \rightarrow 0} \frac{(x+y) \sec(x+y) - x \sec x}{y}$$

10. In the electrolysis of fused salt, the amount of the substance deposited on electrode will not depend on  
a) current intensity  
c) temperature of the bath
11. The oxide which cannot act as a reducing agent is  
a)  $\text{NO}_2$   
b)  $\text{CO}_2$   
c)  $\text{SO}_2$   
d)  $\text{ClO}_2$

18. a) Find the value of 'k' for which the roots of the equation. [2]  
 $x^2 - kx + (k+1) = 0$  are in the ratio 2: 3.

b) Find the condition that the two quadratic equation  
 $ax^2 + bx + c = 0$  ( $a \neq 0$ ) and  $a_1x^2 + b_1x + c_1 = 0$  ( $a_1 \neq 0$ ) have one root common. [3]

19. a) The AM between two numbers exceeds their GM by 2 and GM exceeds the HM by 1.6. Find the two numbers. [2]

b) A rubber Ball is dropped from a Height of 10m. If it rebounds approximately one half the distance after each fall. Use infinite geometric series to approximate the total distance the ball travels before coming to rest. [3]

**GROUP 'C' [3×8=24]**

20. a) Solve the inequality: [2]

$$\frac{x^2 - 2x}{x - 1} \leq 0$$

b) Construct the truth table for compound statement [2]

$$\sim(p \wedge q) \iff (\sim p \vee \sim q).$$

c) Find square root of complex number [2]

$$\frac{5+12i}{3-4i}$$

d) Find the quadratic equation with rational coefficients whose one of the [2]

$$\text{root is } \frac{1}{5+3i}$$

21. a) Find  $\frac{dy}{dx}$  if [1+2+2]

$$\text{i) } y = \frac{f(x)}{g(x)}, \quad (g(x) \neq 0) \quad \text{ii) } y = \sqrt{\frac{x}{a+x}} \quad \text{iii) } x^3 + y^3 - 3xy^2 = 0$$

b) Find differential coefficient of  $\sqrt{\tan 2x}$  by using first principle. [3]

22. A general equation of second degree is given by

$$18x^2 + 6xy - 4y^2 - 15x - 40y - 75 = 0 \dots\dots\dots \text{eq}^n(1)$$

i) Prove that  $\text{eq}^n(1)$  represents a pair of straight lines. [1]

ii) Find separate equation of lines given by  $\text{eq}^n(1)$ . [2]

iii) Find the angle between the lines given by  $\text{eq}^n(1)$ . [1]

iv) Find equation of bisectors of angle between the lines obtained by  $\text{eq}^n(1)$ . [1]

v) Identify the acute angle bisectors. [1]

vi) Identify the bisectors that contain the origin. [1]

*Best of Luck!*

- iv) Find the quadratic equation whose one root is  $-2 + \sqrt{3}$   
 v) Find the value of  $3^{1/3} \cdot 3^{1/9} \cdot 3^{1/27} \dots$

[1]

[1]

19. a) Evaluate:  $\lim_{x \rightarrow \frac{\pi}{2}} \frac{e^{\cos x} - 1}{\frac{\pi}{2} - x}$

[2]

- b) If the quadratic equations  $x^2 + px + q = 0$  and  $x^2 + p'x + q' = 0$  have a common root then prove that the root must be either  $\frac{pq' - p'q}{q - q'}$  or  $\frac{q - q'}{p' - p}$

[3]

### GROUP-B [3×8=24]

20. a) A square is drawn by joining the mid points of the sides of a given square. A third square is drawn inside the second square in the same way and this process continues indefinitely. If a side of the first square is 16 cm, determine the sum of the areas of all the square.

[3]

- b) The sum of an infinite number of terms of a GP is 23 and sum of their square is 69. Determine the first term and the common ratio.

[2]

- c) If the roots of the equation  $ax^2 + cx + c = 0$  are in the ratio m:n, then prove that:  $\sqrt{\frac{m}{n}} + \sqrt{\frac{n}{m}} + \sqrt{\frac{c}{a}} = 0$

[3]

21. a) Define symmetric difference of two sets A & B. For any sets A and B, prove that  $A \Delta B = (A \cup B) - (A \cap B)$

[1+3]

- b) Prove that for any two statements p and q

[2]

i)  $\sim(p \vee q) \equiv \sim p \wedge \sim q$

[2]

ii)  $\sim(p \vee (\sim q)) \equiv (\sim p) \wedge q$

22. Evaluate the following limits:

[2]

a)  $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x}$

[2]

b)  $\lim_{x \rightarrow \infty} \sqrt{x}(\sqrt{x+5} - \sqrt{x})$

[2]

c)  $\lim_{\theta \rightarrow \frac{\pi}{4}} \frac{\cos \theta - \sin \theta}{\theta - \frac{\pi}{4}}$

- d) If  $f(x) = \frac{ax+b}{x-1}$ ,  $\lim_{x \rightarrow 0} f(x) = 2$  and  $\lim_{x \rightarrow \infty} f(x) = 5$ , find the value of  $f(4)$

[2]

Best of luck!

- iv) Find the quadratic equation whose one root is  $-2 + \sqrt{3}$  [1]  
 v) Find the value of  $3^{1/3}, 3^{1/9}, 3^{1/27}, \dots$  [1]  
 19. a) Evaluate:  $\lim_{x \rightarrow \frac{\pi}{2}} \frac{e^{\cos x} - 1}{\frac{\pi}{2} - x}$  [2]  
 b) If the quadratic equations  $x^2 + px + q = 0$  and  $x^2 + p'x + q' = 0$  have a common root then prove that the root must be either  $\frac{pq' - p'q}{q - q'}$  or  $\frac{q - q'}{p' - p}$  [3]

**GROUP-B [3×8=24]**

20. a) A square is drawn by joining the mid points of the sides of a given square. A third square is drawn inside the second square in the same way and this process continues indefinitely. If a side of the first square is 16 cm, determine the sum of the areas of all the square. [3]  
 b) The sum of an infinite number of terms of a GP is 23 and sum of their square is 69. Determine the first term and the common ratio. [2]  
 c) If the roots of the equation  $ax^2 + cx + c = 0$  are in the ratio  $m:n$ , then prove that:  $\sqrt{\frac{m}{n}} + \sqrt{\frac{n}{m}} + \sqrt{\frac{c}{a}} = 0$  [3]
21. a) Define symmetric difference of two sets A & B. For any sets A and B, prove that  $A \Delta B = (A \cup B) - (A \cap B)$  [1+3]  
 b) Prove that for any two statements  $p$  and  $q$  [2]  
 i)  $\sim(p \vee q) \equiv \sim p \wedge \sim q$  [2]  
 ii)  $\sim(p \vee (\sim q)) \equiv (\sim p) \wedge q$
22. Evaluate the following limits: [2]  
 a)  $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x}$  [2]  
 b)  $\lim_{x \rightarrow \infty} \sqrt{x}(\sqrt{x+5} - \sqrt{x})$  [2]  
 c)  $\lim_{\theta \rightarrow \frac{\pi}{4}} \frac{\cos \theta - \sin \theta}{\theta - \frac{\pi}{4}}$   
 d) If  $f(x) = \frac{ax+b}{x-1}$ ,  $\lim_{x \rightarrow 0} f(x) = 2$  and  $\lim_{x \rightarrow \infty} f(x) = 5$ , find the value of  $f(4)$  [2]

*Best of luck!*

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**First Term Examination- 2080**

Class: XI 'Science'  
 Time: 3 hrs.

SET-A  
 Subject: Mathematics [0071]

F.M: 75  
 P.M: 30

Candidates are required to give their answers in their own words as far as practicable.  
 The figures in the margin indicate full marks.

Attempt all the questions.

**GROUP-A [MCQ] [11×1=11]**

- For what value of  $k$ , the equation  $(3k+1)x^2 + 2(k+1)x + k = 0$  may have reciprocal roots?  
 a)  $-\frac{3}{2}$                       b)  $-\frac{1}{2}$                       c)  $\frac{3}{2}$                       d) 2
- If  $A = \{a, b, c, d\}$  then total number of non-empty proper subsets of A is ...  
 a) 4                      b) 14                      c) 15                      d) 16
- Both roots of the equation  $x^2 + ax + b = 0$  and  $x^2 + dx + e = 0$  are common then  
 a)  $ab = de$                       b)  $d^2e = a^2b$                       c)  $ae = bd$                       d)  $ad = be$
- A and B are two non-empty sets then which one is correct?  
 a)  $(A \cup B) \subset (A \cap B)$                       b)  $(A \cap B) \subset (A \cup B)$   
 c)  $(A \cup B) = (A \cap B)$                       d)  $(A \cup B) - (A \cap B) = \emptyset$
- Let 'p' be "10 is an even number" and 'q' be "10 is multiple of 5" then symbolic representation of "10 is neither even nor multiple of 5" is  
 a)  $p \wedge q$                       b)  $(\sim p) \vee (\sim q)$                       c)  $p \vee q$                       d)  $(\sim p) \wedge (\sim q)$
- The value of  $\lim_{x \rightarrow \infty} x \cdot \sin \frac{1}{x}$   
 a) 1                      b) -1                      c) 0                      d) doesn't exist
- $\lim_{x \rightarrow 0} \frac{\ln(x+1)}{e^x - 1} = ?$   
 a) -1                      b) 0                      c) 1                      d) doesn't exist
- The AM and HM between two number are 27 and 12 respectively then GM between two numbers be  
 a) 18                      b) 9                      c) 36                      d) 45
- The value of limit  $\lim_{x \rightarrow 4} \frac{x^2 - 2x - 8}{x - 4}$  is  
 a) 0                      b) 2                      c) 8                      d) 6

- The compound statement  $p \wedge (\sim p)$  is  
 a) tautology                      b) contradiction                      c) implication                      d) inverse
- The sum of the infinite geometric series  $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} \dots$  is  
 a) 0.5                      b) 1                      c) 1.5                      d) 3

**GROUP-B [8×5=40]**

- a) Define subset of a set with example.  
 b) For non-empty sets A, B and C and universal set U, prove that  
 i)  $\overline{A \cap B} = \overline{A} \cup \overline{B}$   
 ii)  $A - (B \cap C) = (A - B) \cup (A - C)$  [1+2+2]
- a) Define tautology and contradiction of statement with example.  
 b) For any statement p and q, test whether  $(p \wedge q) \wedge \sim(p \vee q)$  is a contradiction or not. [2+3]
- a) If one root of the equation  $ax^2 + bx + c = 0$  ( $a \neq 0$ ) be square of the other, then prove that  $b^3 + a^2c + ac^2 = 3abc$  [2]  
 b) Under what conditions the roots of quadratic equation  $ax^2 + bx + c = 0$  ( $a \neq 0$ ) are [2]  
 i) real and unequal                      ii) imaginary                      iii) equal  
 iv) equal in magnitude but opposite in sign  
 c) For what value of k, the equation  $9x^2 + kx + 1 = 0$  has equal roots? [1]
- a) The sum of three numbers in AP is 15. If 1, 4, 19 are added to them respectively, the resulting series is in GP. Find the numbers. [2]  
 b) If a, b, c are in HP, then prove that  $\frac{b+c}{bc}, \frac{a+c}{ac}, \frac{a+b}{ab}$  are in AP. [3]
- a) Define the limit of a function at a point.  
 b) A function given by;  $f(x) = \frac{x^2 - 16}{x - 4}$ . Does f(4) exist? Also find  $\lim_{x \rightarrow 4} f(x)$  (if exist) [2]  
 c) Evaluate:  $\lim_{x \rightarrow 64} \frac{\sqrt[3]{x} - 2}{\sqrt{x} - 4}$  [2]
- Evaluate:  
 a)  $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$                       b)  $\lim_{x \rightarrow y} \frac{x \tan x - y \tan y}{x - y}$  [2+3]
- i) What is the value of  $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a}$  for all rational n and  $n \neq 0$  [1]  
 ii) If  $A \subseteq B$  then show that  $\overline{B} \subseteq \overline{A}$   
 iii) Write the negation of the statement "If 10 is odd or 12 is even, then  $5 > 7$ " [1]

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SET-B

Subject: Mathematics [0071]

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P.M: 30

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Attempt all the questions.

**GROUP-A [MCQ] [11×1=11]**

- The AM and HM between two number are 27 and 12 respectively then GM between two numbers be  
 a) 18                      b) 9                      c) 36                      d) 45
- The sum of infinite geometric series  $\frac{1}{2} + \frac{3}{8} + \frac{9}{32} + \dots$  is  
 a) 1                      b) 2                      c) 4                      d) doesn't exist
- The value of  $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{(\frac{\pi}{2} - x)}$   
 a) 1                      b) -1                      c) 0                      d) doesn't exist
- For what value of  $k$ , the equation  $(3k+1)x^2 + 2(k+1)x + k = 0$  may have reciprocal roots?  
 a)  $-\frac{3}{2}$                       b)  $-\frac{1}{2}$                       c)  $\frac{3}{2}$                       d) 2
- The value of limit  $\lim_{x \rightarrow 4} \frac{\sqrt{x}-2}{x-4}$  is  
 a) 0                      b)  $\frac{1}{4}$                       c)  $\frac{3}{4}$                       d) None
- The compound statement  $p \wedge (\sim p)$  is  
 a) tautology                      b) contradiction                      c) implication                      d) inverse
- Both roots of the equation  $x^2 + ax + b = 0$  and  $x^2 + dx + e = 0$  are common then  
 a)  $ab = de$                       b)  $d^2e = a^2b$                       c)  $ae = bd$                       d)  $ad = be$
- $\lim_{x \rightarrow 0} x \cdot \sin \frac{1}{x}$  is  
 a) 0                      b) 1                      c)  $\infty$                       d) doesn't exist
- Let 'p' be "10 is an even number" and 'q' be "10 is multiple of 5" then symbolic representation of "10 is neither even nor multiple of 5" is  
 a)  $p \wedge q$                       b)  $(\sim p) \vee (\sim q)$                       c)  $p \vee q$                       d)  $(\sim p) \wedge (\sim q)$

**GROUP-B [8×5=40]**

- If  $A = \{a, b, c, d\}$  then total number of non-empty proper subsets of A is ...  
 a) 4                      b) 14                      c) 15                      d) 16
- If A and B are disjoint sets then which of the following is true?  
 a)  $A \subseteq \bar{B}$                       b)  $B \subseteq \bar{A}$   
 c)  $\bar{A} \subseteq \bar{B}$                       d)  $\bar{B} \subseteq \bar{A}$
- Define subset of a set with example.  
 b) For non-empty sets A, B and C and universal set U, prove that  
 i)  $\overline{A \cap B} = \bar{A} \cup \bar{B}$   
 ii)  $\overline{A - (B \cap C)} = (A - B) \cup (A - C)$                       [1+2+2]
- Define tautology and contradiction of statement with example.  
 b) For any statement p and q, test whether  $(p \wedge q) \wedge \sim(p \vee q)$  is a contradiction or not.                      [2+3]
- a) If one root of the equation  $ax^2 + bx + c = 0$  ( $a \neq 0$ ) be square of the other, then prove that  $b^3 + a^2c + ac^2 = 3abc$                       [2]  
 b) Under what conditions the roots of quadratic equation  $ax^2 + bx + c = 0$  ( $a \neq 0$ ) are  
 i) real and unequal                      ii) imaginary                      iii) equal                      iv) equal in magnitude but opposite in sign  
 c) For what value of k, the equation  $9x^2 + kx + 1 = 0$  has equal roots?                      [1]
- a) The sum of three numbers in AP is 15. If 1, 4, 19 are added to them respectively, the resulting series is in GP. Find the numbers.                      [2]  
 b) If a, b, c are in HP, then prove that  $\frac{b+c}{bc}, \frac{a+c}{ac}, \frac{a+b}{ab}$  are in AP.                      [3]
- a) Define the limit of a function at a point.                      [1]  
 b) A function given by;  $f(x) = \frac{x^2-16}{x-4}$ . Does f(4) exist? Also find  $\lim_{x \rightarrow 4} f(x)$  (if exist)                      [2]  
 c) Evaluate:  $\lim_{x \rightarrow 64} \frac{\sqrt[6]{x}-2}{\sqrt[3]{x}-4}$                       [2]
- Evaluate:  
 a)  $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$                       b)  $\lim_{x \rightarrow y} \frac{x \tan x - y \tan y}{x-y}$                       [2+3]
- i) What is the value of  $\lim_{x \rightarrow a} \frac{x^n - a^n}{x-a}$  for all rational n and  $n \neq 0$                       [1]  
 ii) If  $A \subseteq B$  then show that  $\bar{B} \subseteq \bar{A}$                       [1]  
 iii) Write the negation of the statement "If 10 is odd or 12 is even, then  $5 > 7$ "                      [1]

Class: XI 'Science'  
Time: 3 hrs.

Subject: Com. Mathematics (0071)

F.M.: 75

P.M.: 30

Candidates are required to give their answers in their own words as far as practicable.  
The figures in the margin indicate full marks.

Attempt all the questions.

GROUP 'B' [8×5=40]

Short Answer Questions.

12. a) Write the contrapositive of the compound statement  $q \Rightarrow p$ . [1]

b) Write the negation of the statement "If 11 is odd or 2 is even then  $5 < 7$ ." [2]

c) Construct the truth table for the compound statement  $(p \wedge q) \Rightarrow (p \vee q)$ .  
Also, what conclusion can be drawn from the truth table? [2]

13. a) Define power set with an example. [2]

b) If A, B and C are subsets of a universal set U, prove that

i)  $A \subseteq B \Rightarrow \bar{B} \subseteq \bar{A}$ . [1]

ii)  $A \subseteq B \Rightarrow C - B \subseteq C - A$ . [2]

14. a) Find the values of the parameter a for which the quadratic equation

$$(a+1)x^2 + 2(a+1)x + (a-2) = 0$$

i) has two distinct roots [1]

ii) has no real roots [1]

iii) has two equal roots [1]

b) If one root of the equation  $x^2 - \sqrt{5}x - 19 = 0$  is  $\frac{9+\sqrt{5}}{2}$ , then find the other root. [2]

15. a) If  $a^x = b^y = c^z$  and a, b, c are in G.P then prove that x, y and z are in H.P. [3]

b) Under what condition sum of infinite geometric series exists? Give an example and find its sum. [2]

16. a) Under what condition  $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$  for a, b real numbers? Give an example to support this expression. [1]

b) Find the real and imaginary part of the complex number  $\frac{2+\sqrt{-4}}{3+2i}$ . [2]

c) Find the values of following.

i)  $i^{10} + i^{25} + i^{30} + i^{35}$  [1+1=2]

ii)  $(-\sqrt{-1})^{4n+3}, n \in \mathbb{N}$

Evaluate:

a)  $\lim_{x \rightarrow 4} \frac{x^2-16}{\sqrt{3x+4}-4}$  [2+3=5]

b)  $\lim_{x \rightarrow 2} \frac{\sqrt{x-2}+\sqrt{x}-\sqrt{2}}{\sqrt{x^2-4}}$

18. Evaluate:

a)  $\lim_{x \rightarrow 4} \frac{e^x - e^4}{\sqrt{x} - 2}$

b)  $\lim_{h \rightarrow 0} \frac{\sqrt{\sin(x+h)} - \sqrt{\sin x}}{h}$

19. a) Give an example of function which takes the form  $\frac{0}{0}$ , but whose limit does not exist.

b) Is it possible to write  $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = \lim_{x \rightarrow 0} x^2 \cdot \lim_{x \rightarrow 0} \sin \frac{1}{x}$ ? If not, give reason.

c) Evaluate:

i)  $\lim_{x \rightarrow \infty} \frac{3x}{\sqrt{x^2+5}}$

GROUP 'C' (3×8=24)

20. a) Under what condition two compound statements are logically equivalent? Justify it by an example.

b) Define Set. How does it differ from collection? Justify it by an example.

c) If A, B and C are subsets of universal set U, prove that.

i)  $\overline{(A \cup B)} = \bar{A} \cap \bar{B}$

ii)  $A - (B \cup C) = (A - B) \cap (A - C)$

21. a) Define identity. How does it differ from an equation?

b) Under what condition roots of quadratic equation  $px^2 + qx + r = 0$  are imaginary? Justify it by an example.

c) Derive the condition for two quadratic equations  $p_1x^2 + q_1x + r_1 = 0$  and  $p_2x^2 + q_2x + r_2 = 0$  to have one root common.

d) If each term of an infinite geometric series is equal to three times the sum of all the terms which follow it and the sum of first two terms is 1. Find the sum to infinity of the series.

22. a) Evaluate:

i)  $(1+i)^6 \left(1+\frac{1}{i}\right)^6$

ii)  $\lim_{x \rightarrow y} \frac{x \sin x - y \sin y}{x-y}$

b) If a, b, c are in A.P; b, c, a are in H.P then prove that c, a, b are in G.P.

c) For a given function  $f(x) = \frac{x-4}{\sqrt{x}-2}$ .

i) Is it possible to find f(2)? Explain.

ii) Does  $\lim_{x \rightarrow 4} f(x)$  exist? Explain.

SET-A

Science'

Mins.

Subject: Chemistry [3011]

Best options among the four options. Apply tick on two. A mark will not be counted.

Choice Questions.

GROUP 'A' [MCQ] [11×1=11]

Rate of diffusion of which of the following gases is fastest?

b) H<sub>2</sub> c) NO<sub>2</sub>

Density of NH<sub>3</sub> at 30°C and 5 atm pressure is

a) 30 gm/lit b) 2.43 gm/lit c) 3.42 gm/lit

Atmospheric air contains 20% oxygen by volume, p

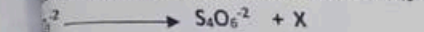
a) 2 mmHg b) 760 mmHg c) 380 mmHg

Increase in volume that will take place when

gas is heated to 360K at constant pressure is

a) 10 ml b) 200 ml c) 160 ml

In the following equation, the value of 'X' is



a) 3e<sup>-</sup> b) 3e<sup>+</sup> c) 4e<sup>-</sup>

In the electrolysis of fused salt, the amount of

metal deposited at the cathode will not depend on

a) current intensity b) time

c) temperature of the both d) electrode

Which of the following oxides cannot act as a reducing agent?

a) CO b) CO<sub>2</sub> c) SO<sub>2</sub>

Number of moles of Faraday needed to deposit 1 mole of

a) 1 b) 2 c) 3

Most stable allotrope of sulphur is

a) rhombic b) plastic c) monoclinic

Correct order of size is

a) I<sup>+</sup> < I<sup>+</sup> < I<sup>+</sup> b) I<sup>+</sup> < I<sup>+</sup> < I<sup>+</sup> c) I<sup>+</sup> < I<sup>+</sup> < I<sup>+</sup>

Correct order of ionization energy is

a) N > O > C b) N > O > C c) O > N > C

Class: XI 'Science'

SET-B

Time: 25 min.

Subject: Com. Mathematics (0071)

Tick the best options among the four options. Apply tick on two options will be automatically deslaid and mark will not be counted.

GROUP 'A' MCQ [11×1=11]

Re-write the correct option of each question in your answer sheet.

1. What is the sum of the series  $9 + 3 + 1 + \frac{1}{3} + \dots$  ?

A)  $\frac{13}{2}$  B) 13 C)  $\frac{25}{2}$  D)  $\frac{27}{2}$

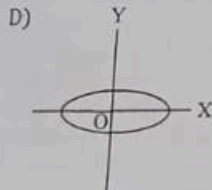
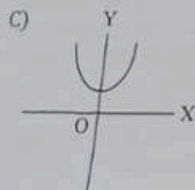
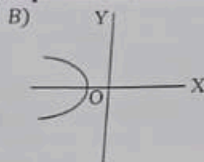
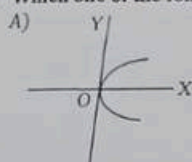
2. Which of the following compound statement is not true?

A)  $\sim(p \wedge q) \Leftrightarrow [(\sim p) \vee (\sim q)]$  B)  $\sim(p \vee q) \Leftrightarrow [(\sim p) \wedge (\sim q)]$   
C)  $\sim(p \Rightarrow q) \Leftrightarrow [p \wedge \sim q]$  D)  $p \wedge (q \vee r) \Leftrightarrow [(p \vee q) \wedge (p \wedge r)]$

3. What is the solution of  $|2 - x| \geq 5$ ?

A)  $(-\infty, \infty)$  B)  $(3, 7)$   
C)  $(-\infty, -3) \cup (7, \infty)$  D)  $(-\infty, -3] \cup [7, \infty)$

3. Which one of the following graphs represents a function?



5. What is the multiplicative inverse of  $z = (3, -2)$ ?

A)  $\frac{3}{13} - \frac{2}{13}i$  B)  $\frac{-3}{13} + \frac{2}{13}i$  C)  $\frac{-3}{13} - \frac{2}{13}i$  D)  $\frac{3}{13} + \frac{2}{13}i$

6. What is the length of perpendicular from the point  $(b, a)$  to the line  $\frac{x}{a} + \frac{y}{b} = 1$ ?

A)  $\frac{b^2 - a^2 - ab}{\sqrt{a^2 + b^2}}$  B)  $\frac{b^2 + a^2 + ab}{\sqrt{a^2 - b^2}}$  C)  $\frac{b^2 + a^2 - ab}{\sqrt{a^2 + b^2}}$  D)  $\frac{b^2 - ab + b^2}{\sqrt{b^2 - a^2}}$

7. What is the angle between the line pair represented by  $2x^2 + xy - y^2 = 0$  ?

A)  $\theta = \tan^{-1}|\sqrt{3}|$

B)  $\tan^{-1}|\frac{1}{3}|$

C)  $\tan^{-1}|3|$

D)  $\tan^{-1}|\frac{1}{\sqrt{3}}|$

8. The roots of the equation  $x^2 + 2\sqrt{3}x + 3 = 0$  are

A) real and unequal

B) rational and equal

C) irrational and unequal

D) irrational and equal

9. What is the value of  $\lim_{x \rightarrow 2^+} \frac{\sin(x-2)}{|2-x|}$  ?

A) 1

B) -1

C) 0

D) does not exist

10. What is the value of k, if the function  $f(x) = \begin{cases} x & \text{for } x \neq 0 \\ k & \text{for } x = 0 \end{cases}$  is

continuous at  $x = 0$ ?

A)  $\frac{1}{\sqrt{3}}$

B)  $\sqrt{3}$

C) 3

D)  $\frac{1}{3}$

11. What is the derivative of  $2(\sqrt{x} - \frac{1}{\sqrt{x}})$ ?

A)  $\frac{x-1}{2x\sqrt{x}}$

B)  $\frac{x-1}{x\sqrt{x}}$

C)  $\frac{x+1}{2x\sqrt{x}}$

D)  $\frac{x+1}{x\sqrt{x}}$

XI 'Science'

Subject: Com. Math

Best options among the four options. Apply tick on two options will not be counted.

Attempt all the questions.

GROUP 'B'

Construct a truth table for the statement  $\sim(p \vee q)$ .

State and prove De-Morgan's law.

Solve the inequality  $\frac{x^2 - 2x}{x - 3} \geq 0$ .

Also, show the solution in the number line.

Under what condition inverse of a function  $f: R \rightarrow R$  defined by  $f(x) = \begin{cases} x & \text{if } x \text{ is rational} \\ x+1 & \text{if } x \text{ is irrational} \end{cases}$  exists?

Can a positive number between 0 and 1 be written as a sum of two positive numbers?

Justify it by an example.

If a, b, c are in A.P., a, x, b and b, x, c are in G.P., find the value of  $\frac{a}{b} \cdot \frac{b}{c} \cdot \frac{c}{a}$ .

Write any two differences between real and complex numbers.

If  $z = x + yi = \sqrt{\frac{1+i}{1-i}}$ , show that  $|z| = 1$ .

Under what condition a polynomial is divisible by another polynomial?

Imaginary? Justify it by an example.

Find the condition that the roots of the equation  $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$  are real.

If  $A = (-1, 4)$  and  $B = (1, -4)$ , find the equation of the line segment AB.

Find  $\frac{dy}{dx}$  when  $\sqrt{x^2 + y^2} = \sqrt{1 + x^2}$ .

Evaluate:  $\lim_{x \rightarrow a} \frac{\sin 2x - \sin 2a}{x - a}$ .

Evaluate each of the following limits.

values to confirm your answer.

$\lim_{x \rightarrow 0} \frac{1}{x}$  ii)  $\lim_{x \rightarrow \infty} \frac{1}{x}$

iii)  $\lim_{x \rightarrow 0} \frac{6x^2 + x - 2}{2x - 1}$

iv)  $\lim_{x \rightarrow 0} \frac{5}{2}$

Not continuous at  $x = 0$ .

Science

Subject: Com. Mathematics (0071)

F.M.: 75

P.M.: 30

Options among the four options. Apply tick on two options will be automatically denied

all the questions.

**GROUP 'B' [8×5=40]**

Construct a truth table for the statement  $\sim(p \vee (\sim q))$ .

State and prove De-Morgan's laws for two sets.

Solve the inequality  $\frac{x^2-2x}{x-3} \geq 0$ .

Also, show the solution in the number line.

What condition inverse of  $f(x)$  exists? Prove that the function

$f: \{0\} \rightarrow \mathbb{R} - \{2\}$  defined by  $f(x) = \frac{2x+1}{x}$  is bijective. Also, find  $f^{-1}$  (if exists).

Find a positive number between two negative numbers be geometric mean?

Justify it by an example.

If  $a, b, c$  are in A.P.,  $a, x, b$  and  $b, y, c$  are in G.P. Show that  $x^2, b^2, y^2$  are in A.P.

State any two differences between real numbers and complex numbers.

If  $z = x + yi = \sqrt{\frac{1+i}{1-i}}$ , show that  $x^2 + y^2 = 1$ . Also, find  $\bar{z}$ .

Under what condition a pair of lines represented by  $ax^2 + 2hxy + by^2 = 0$  are

imaginary? Justify it by an example.

Under the condition that the general equation of second degree

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$  represents a pair of lines.

If  $A = (-1, 4]$  and  $B = [1, 3]$ , find  $A \cap B$  and  $A - B$ .

Find  $\frac{dy}{dx}$  when  $\sqrt[3]{x^2} + \sqrt[3]{y^2} = \sqrt[3]{a^2}$

Evaluate:  $\lim_{x \rightarrow a} \frac{\sin 2x - \sin 2a}{\sqrt{x} - \sqrt{a}}$

Evaluate each of the following limits, if possible. Use a table of functional limits to confirm your conclusion.

i)  $\lim_{x \rightarrow 0^-} \frac{1}{x}$

ii)  $\lim_{x \rightarrow 0^+} \frac{1}{x}$

iii)  $\lim_{x \rightarrow 0} \frac{1}{x}$

**GROUP 'C' [3×8=24]**

If  $f(x) = \begin{cases} \frac{6x^2+x-2}{2x-1} & \text{if } x = \frac{1}{2} \\ \frac{5}{2} & \text{if } x \neq \frac{1}{2} \end{cases}$ , decide whether  $f$  is continuous at  $x = \frac{1}{2}$ . If

$f$  is not continuous at  $x = \frac{1}{2}$ , what type of discontinuity is it?

b) Define derivative of the function  $f(x)$  with respect to  $x$ . By using definition, find the derivative of  $\sqrt{\tan 5x}$

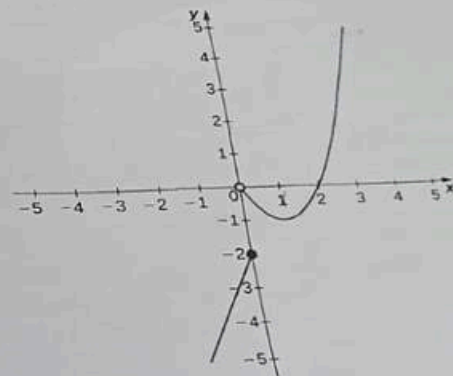
21. a) Find the domain and range of the function  $f(x) = \sqrt{x^2 + x - 6}$

b) Prove that the straight lines joining the origin to the points of intersection of the line  $\frac{x}{a} + \frac{y}{b} = 1$  and the curve  $x^2 + y^2 = c^2$  are at right angles

if  $\frac{1}{a^2} + \frac{1}{b^2} = \frac{2}{c^2}$ .

22. a) If the equation  $x^2 - px + q = 0$  and  $x^2 - ax + b = 0$  have a common root and the other root of the second equation is the reciprocal of the other root of the first, then prove that  $(q-b)^2 = bq(p-a)$ .

b) Using the given graph of the function  $y = f(x)$ , find the answers to the following questions.



i)  $\lim_{x \rightarrow 0^-} f(x)$  and  $\lim_{x \rightarrow 0^+} f(x)$

ii)  $\lim_{x \rightarrow 0} f(x)$

iii)  $f(0)$

iv) Is  $f(x)$  continuous at  $x = 0$ ? If not continuous, what type of discontinuity is it?

v) Range of  $f(x)$

"Best of Luck"

**KATHMANDU MODEL SECONDARY SCHOOL**  
**SECOND TERM EXAMINATION-2081**  
**SET-A**

**Subject: Com. Mathematics (0071)**

Science'

Options among the four options. Apply tick on two options will be automatically denied

**GROUP 'A' MCQ [11×1=11]**

the correct option of each question in your answer sheet.

of the following compound statement is not true?

A)  $(p \wedge q) \Leftrightarrow [(\sim p) \vee (\sim q)]$

B)  $\sim(p \vee q) \Leftrightarrow [(\sim p) \wedge (\sim q)]$

C)  $(p \Rightarrow q) \Leftrightarrow [p \vee \sim q]$

D)  $p \wedge (q \vee r) \Leftrightarrow [(p \wedge q) \vee (p \wedge r)]$

is the solution of  $|1 - x| > 5$ ?

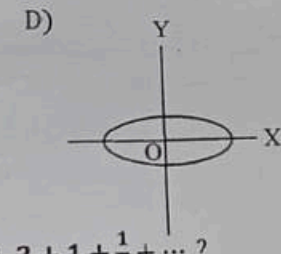
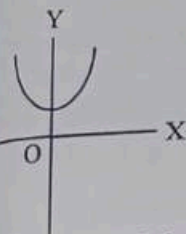
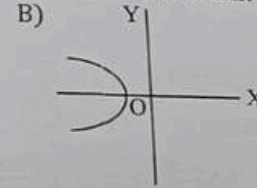
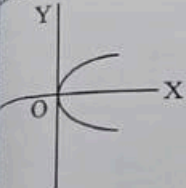
A)  $(-\infty, \infty)$

B)  $(4, 6)$

C)  $(-\infty, -4) \cup (6, \infty)$

D)  $(-\infty, 4] \cup [6, \infty)$

one of the following graph represents a function?



is the sum of the series  $9 + 3 + 1 + \frac{1}{3} + \dots$ ?

A)  $13\frac{1}{2}$

B)  $\frac{25}{2}$

C)  $\frac{29}{2}$

is the multiplicative inverse of  $z = (-3, 2)$ ?

A)  $-\frac{2}{13}i$

B)  $\frac{-3}{13} + \frac{2}{13}i$

C)  $\frac{-3}{13} - \frac{2}{13}i$

D)  $\frac{-3}{13} - \frac{1}{13}i$

is the length of perpendicular from the point  $(b, a)$  to the

A)  $-\frac{y}{b} = 1$ ?

B)  $\frac{b^2 - a^2 + ab}{\sqrt{a^2 - b^2}}$

C)  $\frac{b^2 + a^2 - ab}{\sqrt{a^2 + b^2}}$

D)  $\frac{b^2 - ab + b^2}{\sqrt{b^2 - a^2}}$

is the angle between the line pair represented by  $2x^2 - xy - y^2 = 0$ ?

A)  $\theta = \tan^{-1}|\sqrt{3}|$

B)  $\tan^{-1}|\frac{1}{\sqrt{3}}|$

C)  $\tan^{-1}|3|$

D)  $\tan^{-1}|\frac{1}{3}|$

8. The roots of the equation  $x^2 + 2\sqrt{3}x + 3 = 0$  are

A) real and unequal

B) rational and equal

C) irrational and equal

D) irrational and unequal

9. What is the value of  $\lim_{x \rightarrow 1} \frac{\sin(x-1)}{|1-x|}$ ?

A) 1

B) -1

C) 0

D) does not exist

10. What is the value of k, if the function  $f(x) = \begin{cases} \frac{x}{e^{2x}-1} & \text{for } x \neq 0 \\ k & \text{for } x = 0 \end{cases}$  is continuous at  $x = 0$ ?

A)  $\frac{1}{\sqrt{2}}$

B)  $\sqrt{2}$

C) 2

D)  $\frac{1}{2}$

11. What is the derivative of  $2(\sqrt{x} + \frac{1}{\sqrt{x}})$ ?

A)  $\frac{x-1}{2x\sqrt{x}}$

B)  $\frac{x-1}{x\sqrt{x}}$

C)  $\frac{x+1}{2x\sqrt{x}}$

D)  $\frac{x+1}{x\sqrt{x}}$

वैज्ञानिक रूपमा पनि मानसिक रोग लाग्ने सम्भावना देखिन्छ। मानसिक रोगको एउटा प्रमुख कारण पारिवारिक भगडा, बेरोजारी, प्राकृतिक प्रकोप भएको हुनाले विरामी हुन नदिन र विरामी भएकालाई छिटो निको पार्ने पारिवारिक सदस्यको भूमिका ठूलो हुन्छ। परिवार सामु देखिने जटिल समस्या एक आपसमा मिलेर समाधान गर्ने सकिने हुनाले उक्त बोझ एउटामा मात्र पर्न दिनु हुदैन। विरामीलाई डाक्टरको सल्लाह बमोजिम नियमित औषधि सेवन गराउनुपर्छ। विरामी रिसाउने, फाँकिने तथा अन्य भगडानु स्वभाव देखाउन बालेमा आफू उत्तेजित हुनुहुदैन। विरामीले रोगको कारणले त्यस्तो व्यवहार देखाएको हो भन्ने बुझ्नुपर्छ। औषधिले काम गर्न कमिनामा एक हप्ता लाग्छ, त्यसैले तत्काल औषधिको अन्तर खोल्नु हुदैन। नेपालमा रक्सी र लागुऔषधको समस्या बढ्दै गएको छ। यसका लागि पारिवारिक सदस्यको रेखरेख र निगरानी अति आवश्यक छ। साथै समुदायमा बस्ने मानिस एकजुट भई काम गरे यो नियन्त्रण गर्न सकिन्छ।

प्रश्नहरू :

क) विराममा कुनै न कुनै प्रकारको मनोरोगबाट पीडित भएका मानिसको सङ्ख्या कति छ ?  
 अ) करीब २० करोड आ) करीब २५ करोड इ) करीब ४ करोड ई) करीब ५० करोड

ख) विराममा कतिवटा खतरनाक रोगले मानिसको दैनिक क्रियाकलापमा असर गर्छ ?  
 अ) ५ वटा आ) १५ वटा इ) १० वटा ई) २० वटा

ग) मानसिक रोग लाग्नुका प्रमुख कारण के हुन् ?  
 अ) पारिवारिक असन्तुष्टि, बेरोजारी, प्राकृतिक असन्तुलन  
 आ) पारिवारिक चिन्ता, अलसता  
 इ) पारिवारिक बेमेल, बेरोजारी, जलवायु परिवर्तन  
 ई) पारिवारिक भगडा, बेरोजारी, प्राकृतिक प्रकोप

घ) मानसिक विरामी भएकालाई छिटो निको पार्न कसको भूमिका ठूलो हुन्छ ?  
 अ) परिवारको आ) श्रीमान र श्रीमतीको  
 इ) छोरा र छोरीको ई) आफ्नै

१०. दिइएको अनुच्छेदबाट चारकोटा बुँदा टिपी एक तृतीयांशमा सारांश लेख्नुहोस्: २+२=४  
 नेपालको भूमिमा प्रशस्त खनिज सम्पदाहरू छन्। नेपाली भूगोलको सतह तथा भूगर्भभित्र सुन, चाँदी, फलाम, तामा, अभ्रक, सिसा, जस्ता, खरी, स्लेट, मार्बल, गेरु, कोइला, ग्याँस, पेट्रोलियम, युरेनियम, सोडा, बिरेनियम आदि अनेकौं खनिजहरू छन्। सुनकोसी, बुढी गण्डकी, काली गण्डकी, रिउ खोला, कोसी आदिमा सुन पाइन्छ। यसका साथै कर्नालीको बगरमा पनि सुन पाइन्छ। चिसापानी गढी, बागलुङ र फूलचौकीमा चाँदी पाइन्छ भने ठोसे, फूलचौकी, धितवन, कुलेखानी, भैँसे, फर्पिङ, डोटी आदिमा प्रशस्त फलाम पाइन्छ। तामाछानी, कुलेखानी, बुढ खोला, बाप्सा, मार्फा, इलाम, म्याग्दी आदिमा तामा र भोजपुर, सिन्धुली, लमजुङ, गोसाइँकुण्ड, सुन्दरीजल जस्ता ठाउँमा अभ्रक भन्ने खनिज पाइन्छ। चोभार, भैँसे, धनकुटा, मोरङ जस्ता ठाउँमा चुनचुडा र गणेश हिमाल, फूलचौकी, मकवानपुर, माजेर खोला आदि ठाउँमा सिसा जस्ता पाइन्छन्। त्यसै गरी चिसापानी गढी, गोसाइँकुण्डमा गन्धक तथा खोटाङ, सिन्धुपाल्चोक, भोजपुरमा दलैख, प्युठान, कोइलाबास, दाङलगायतका धेरै ठाउँमा पेट्रोलियम पाइने सम्भावना छ। अत्यन्त बहुमूल्य मानिने युरेनियम मुक्तिनाथलगायतका केही ठाउँमा पाइने कुरा अनुसन्धानहरूले देखाएको छ। यस्ता खनिजहरूको आधुनिक तरिकाले उत्खनन, प्रशोधन र प्रयोग गर्न आवश्यक छ।

११. कुनै एक प्रश्नको उत्तर दिनुहोस्:

क) धमला वस्त्रालय, वानेश्वरले घिन्त्याण्ड वस्त्रालय, न्युरोडलाई दशैँका लागि आवश्यक पर्ने विभिन्न डिजाइनका कपडाहरू पठाईदिन आग्रह गर्दै लेखिने व्यावसायिक पत्रको नमुना लेख्नुहोस्।

ख) विद्युतीय कारको बजार प्रवर्धनका लागि बनाइने विज्ञापनको सचित्र आकर्षक नमुना बनाउनुहोस्।

१२. कुनै एक प्रश्नको उत्तर दिनुहोस्:

क) केएमसी विद्यालयले आफ्नो पञ्जीसी वार्षिक उत्सव कार्यक्रममा आयोजना गरेको 'मिडिया प्रतियोगिता' का सम्पूर्ण कार्यक्रमलाई समेटेर १५० शब्दसम्मको प्रतिवेदन लेख्नुहोस्।  
 ख) 'मूल्यवृद्धि'ले निम्त्याएका समस्याहरू विषयमा १५० शब्दसम्मको टिप्पणी लेख्नुहोस्।

१३. छउपडी प्रथा बन्त्य गर्न के गर्नुपर्छ ? १२५ शब्दसम्मको अनुच्छेद लेख्नुहोस्:

१४. तलका कुनै दुई प्रश्नको उत्तर दिनुहोस्:

क) दिइएको कवितामा पढी सोधिएका प्रश्नको उत्तर दिनुहोस्:  
 कुन पहाडले कुन खोलाले तिमी गाँत छेकेछ्यो र  
 वीर पुर्खा ? कुन ओधीले तिमी यात्रा रोकेछ्यो र  
 गरुडको भैंँ वेग तिमी कुन आकाशले बाँध्न सक्थ्यो  
 पौरखले रच्यो नेपाल ! पहाड तराई जुट्न सक्थ्यो।

प्रश्नहरू:

क) नेपालीहरूको कस्तो पौरखले नेपालको रचना भएको हो ?  
 ख) कविताशका आधारमा हाम्रो वीर पुर्खाको गौरव गाथाको वर्णन गर्नुहोस्।

ख) दिइएको कविता पढी सोधिएका प्रश्नहरूको उत्तर लेख्नुहोस्:

खोला अचल फेरि गाउँतिर सोधिएको रहेछ। यो खोलाले पनि गाउँलाई धुक्के रुवायो। हुन नसकेको माझैँ भन्नु खोलापति अन्याय छल्लो। यो खोलाले गाउँलाई पालेको पनि छ। हाम्रो विपत्तिको लक्षण आधा उमेर यही खोलाको ढिल र बगरहरूमा बितेको थियो। कहिले ओरहा खाने निर्दोष त छँदै थियो गाउँभैँसी चराउने निर्दोष। खोलामा बाढी आएको बेला हामी गाईको पुच्छर समाएर बाढी तर्ने गर्थौँ। अबचा यस्तो बेलामा हामी खोलाको किनारमा बसेर बाढीको आनन्द लुट्थौँ। तर्ने डराउने केही पाँचौँ आएका भने हामी तीनचार जना एकअर्काको हात समाउँदै बाढीमा हेलिन्थ्यौँ र किनारमा पस्नबाट मानिसलाई तारिदिन्थ्यौँ र यस्तो काममा खास गरी परिछन्नाकै विशेष रुचि थियो। "यहीनिर कुन छ, यो सालको वर्षामा मूल फुटेको एकदम चिसो र मिठो पानी छ।" ऊ पानी लिएर आइसकेको थियो।

प्रश्नहरू:

अ) खोलाले रुवायो भन्ने कुरा मात्र गर्दा कसरी अन्याय हुन्छ ?  
 आ) 'म' पात्रका खोलासँगका बालअनुभूति कस्ता छन् ?

ग) दिइएको चिठीको अंश पढी प्रश्नहरूको उत्तर दिनुहोस्:

मित्र सुरज ! अवश्य नै पढाइ छोडेर धन कमाउन परदेशिने मेरो निर्णय अत्यन्त गलत थियो। मेरो कमाउन नसके पनि आफ्नो गाउँ, आफ्नो परिवार, इष्टमित्र, समाज र देश सम्भेर घर फर्कने मेरो निर्णय गलत हुने छैन। त्यसो त निर्णय आफैँमा सही वा गलत हुँदैन। त्यसलाई सही वा गलत बनाउने त हामी आफैँले हो। त्यसैले घर फर्कने मेरो यो निर्णयलाई म जसरी पनि सही साबित गर्छु।

प्रश्नहरू :

क) म पात्रले गरेको सही र गलत निर्णय के के थिए ?  
 ख) म पात्रको घर फर्कने निर्णयप्रति तपाईंको धारणा लेख्नुहोस्।

१५. तलका मध्ये कुनै एक प्रश्नको उत्तर दिनुहोस्।

क) योगमायाले धर्मराज्यको स्थापनाको लागि राखेका मागहरू केके हुन् र ती मागहरू अहिलेको समयमा कतिको सान्दर्भिक छन् ? योगमायाको जीवनीका आधारमा समीक्षा गर्नुहोस्।  
 ख) 'त्यो फेरि फर्कला ?' कथाका कथाकी मुख्य पात्र सानीको केकस्तो मनोविज्ञानको चित्रण गरेको छ ? 'त्यो फेरि फर्कला ?' कथाका आधारमा समीक्षा गर्नुहोस्।

१६. तल दिइएका मध्ये कुनै एक शीर्षकमा २५० शब्दसम्ममा नघटाई निबन्ध लेख्नुहोस्।

क) मभिन्नको देशप्रेम  
 ग) सामाजिक जागरणमा शिक्षा र सञ्चार

Subject: Com. Mathematics (0071)  
 Time: 5 min.  
 Note: Options among the four options. Apply tick on two options will not be counted.

GROUP 'A' MCQ [11×1=11]

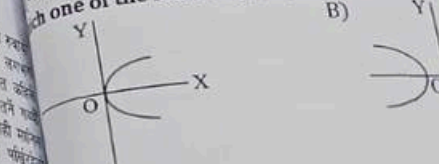
Choose the correct option of each question in your answer.

1. Which of the following compound statement is not true?  
 A)  $(p \wedge q) \Rightarrow [(p \vee q) \wedge (p \wedge q)]$   
 B)  $\sim(p \vee q) \Leftrightarrow [(\sim p) \wedge (\sim q)]$   
 C)  $(p \Rightarrow q) \Leftrightarrow [p \vee \sim q]$   
 D)  $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$

2. What is the solution of  $|1 - x| > 5$ ?

A)  $(-\infty, -4) \cup (6, \infty)$   
 B)  $(-4, 6)$   
 C)  $(-\infty, 4) \cup (6, \infty)$   
 D)  $(-\infty, 4) \cup (6, \infty)$

3. Which one of the following graph represents a function?



4. The sum of the series  $9 + 3 + 1 + \dots$  is

A)  $13\frac{1}{2}$   
 B)  $13$   
 C)  $13\frac{1}{2}$   
 D)  $13$

5. What is the multiplicative inverse of  $\frac{2}{3}$ ?

A)  $\frac{3}{2}$   
 B)  $\frac{-3}{13} + \frac{2}{13}i$   
 C)  $\frac{2}{3}$   
 D)  $\frac{-3}{13} + \frac{2}{13}i$

6. What is the length of perpendicular from the origin to the line  $\frac{x}{a} + \frac{y}{b} = 1$ ?

A)  $\frac{1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}}$   
 B)  $\frac{b^2 - a^2 + ab}{\sqrt{a^2 + b^2}}$   
 C)  $\frac{a^2 - ab}{\sqrt{a^2 + b^2}}$   
 D)  $\frac{b^2 - a^2 + ab}{\sqrt{a^2 + b^2}}$

7. What is the angle between the lines  $y = mx + c$  and  $y = nx + d$ ?

A)  $\tan^{-1} \left| \frac{m-n}{1+mn} \right|$   
 B)  $\tan^{-1} \left| \frac{m+n}{1-mn} \right|$   
 C)  $\tan^{-1} \left| \frac{m-n}{1-mn} \right|$   
 D)  $\tan^{-1} \left| \frac{m+n}{1+mn} \right|$